**Topic 9 of Personnel Economics: Teams**

**Exercises**

## Exercise 9.1

Consider two self-employed workers. If they work alone, each worker's output is given by:

$Q = e_i$

If they work together, the team's output is given by:

$Q = \beta(e_1 + e_2)$

where $e_1$ is effort of worker 1 and $e_2$ is effort of worker 2.

Worker's cost of effort is $\frac{1}{2}\Theta e^2$. The value of output is p per unit. Each worker's utility is equal to his income minus his cost of effort.

a) What is the level of effort, income, and utility of the workers if they work alone?

Assume that when workers decide to work together, they share the value of joint output equally.

b) What is the level of effort, income, and utility of workers if they work together?

c) Under what condition will workers decide to work together?

Suppose a social planner who aims to maximize the sum of the workers' utilities could choose both workers' efforts as well as whether they work together.

d) What is the level of effort, income, and utility of workers if they work alone and if they work together? Under what condition will workers work together?

A social scientist argues that team production works better if team members care about one another. He therefore sees a productivity-enhancing role for social activities like going out for dinner and so on.

e) How would your results of the first part of b) change if workers care about each other's utility?

**Answers**

**9.1**

a) Worker's income equals pQ = pe. Utility maximization results in e = p/Θ, and so w = pe = $p^2/\Theta$. Utility = ½ $p^2/\Theta$.

b) Worker 1's effort choice: max ½ p β($e_1$ + $e_2$) - ½Θ $e_1^2$ results in $e_1$ = ½ β p/Θ. Hence, the wage w = ½ $(\beta p)^2/\Theta$. Utility = 3/8 $(\beta p)^2/\Theta$.

c) Utility when working alone was derived under part a) of this exercise. Utility when working together was derived under part b) of this exercise. Work together if $\beta > \sqrt{4/3} \approx 1{,}15$.

d) The social planner maximizes joint utility = p β($e_1$ + $e_2$) - ½Θ $e_1^2$- ½ Θ $e_2^2$ . Resulting effort levels: $e_i$ = β p/Θ. Wage w = $(\beta p)^2/\Theta$. Utility = ½ $(\beta p)^2/\Theta$. Work together if β>1.

e) Worker 1's effort choice becomes: max ½ p β($e_1$ + $e_2$) - ½Θ $e_1^2$ + ρ[½ p β($e_1$ + $e_2$) - ½Θ $e_2^2$], where ρ measures how much worker 1 cares about worker 2. Implying $e_1$ = ½(1+ ρ) β p/Θ. So the higher is ρ, the higher is effort. If ρ=1, the effort equals socially optimal level of effort.